

ImageNet Classification with Deep Convolutional Neural Networks

Michael Schatz
Oct 1, 2025

JHU EN.601.449/EN.601.649: Applied Comparative Genomics

Assignment 3 Due Wednesday Oct 1 by 11:59pm

The screenshot shows a web browser displaying a GitHub repository page. The browser's address bar shows the URL: `github.com/schatzlab/appliedgenomics2025/blob/main/assignments/assignment3/README.md`. The page has a sidebar on the left with a 'Files' section showing a directory tree. The main content area has tabs for 'Preview', 'Code', and 'Blame'. The 'Preview' tab is active, showing the README content. The README title is 'Assignment 3: BWT and Variant Calling'. Below the title, it lists the assignment date (Wednesday, September 24, 2025) and the due date (Wednesday, October 1, 2025 @ 11:59pm). There is an 'Assignment Overview' section with a paragraph about implementing the BWT and exploring requirements for variant calling. It mentions that programming exercises can be computed in any language, but recommends Python, C++, Java, or Rust, and notes that R is inefficient for string processing. A reminder to post questions on Piazza is included. The 'Question 1. BWT Encoding [20 pts]' section follows, with a paragraph about implementing a BWT encoder and a note about time and space complexity. A recommended pseudo-code snippet is provided in a light blue box.

Files

- main
- Go to file
- assignments
 - assignment1
 - assignment2
 - assignment3
 - input_files
 - README.md
- lectures
- policies
 - .gitignore
 - LICENSE
 - README.md

Assignment 3: BWT and Variant Calling

Assignment Date: Wednesday, September 24, 2025
Due Date: Wednesday, October 1, 2025 @ 11:59pm

Assignment Overview

In this assignment you will implement the BWT and explore the requirements for variant calling. The programming exercises can be computed in any programming language, although we recommend python (or C++, Java, or Rust). R is generally inefficient at string processing unless you take great care. See the resources at the bottom of the page for tips for the variant calling exercises.

As a reminder, any questions about the assignment should be posted to [Piazza](#).

Question 1. BWT Encoding [20 pts]

In the language of your choice, implement a BWT encoder and encode the string below. Faster (Linear time) methods exist for computing the BWT, although for this assignment you can use the simple method based on standard sorting techniques. Your solution does *not* need to be an optimal algorithm and can use $O(n^2)$ space and $O(n^2 \lg n)$ time.

Here is the recommended pseudo code (make sure to submit your code as well as the encoded string):

```
computeBWT(string s)
  ## add the magic end-of-string character
  s = s + "$"

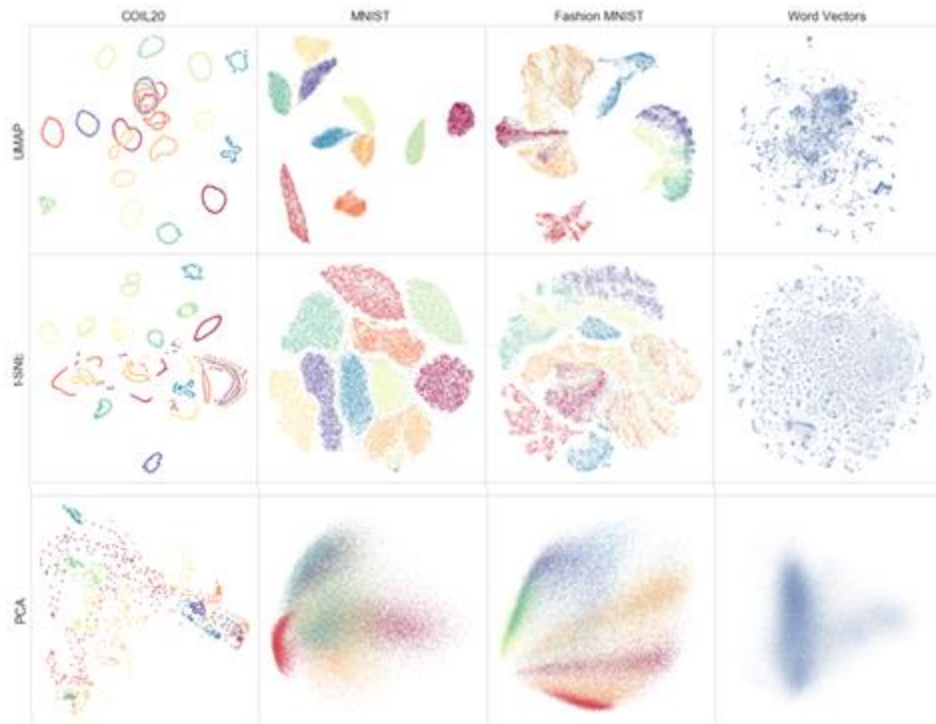
  ## build up the BWM from the cyclic permutations
  ## note the ith cyclic permutation is just "s[i..n] + s[0..i]"
  rows = []
  for (i = 0; i < length(s); i++)
    rows.append(cyclic_permutation(s, i))

  ## just use the builtin sort command to sort the cyclic permutations
  sort(rows)

  ## now extract the last column
  bwt = ""
  for (i = 0; i < length(s); i++)
    bwt += substr(rows[i], length(s), 1)
```

Recap

Dimensionality Reduction



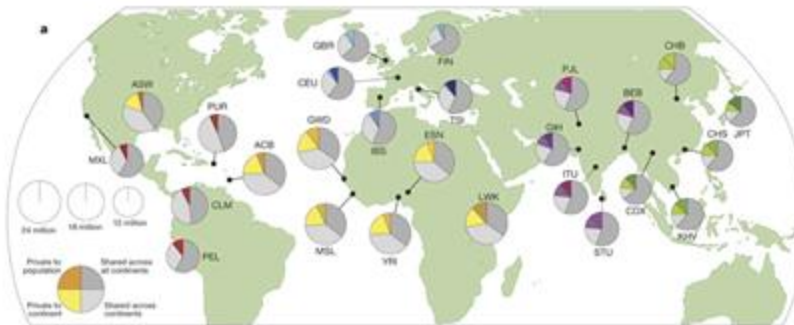
UMAP: Uniform Manifold Approximation and Projection for Dimension Reduction

McInnes et al (2018) arXiv. 1802.03426

<https://www.youtube.com/watch?v=nq6iPZVUxZU>

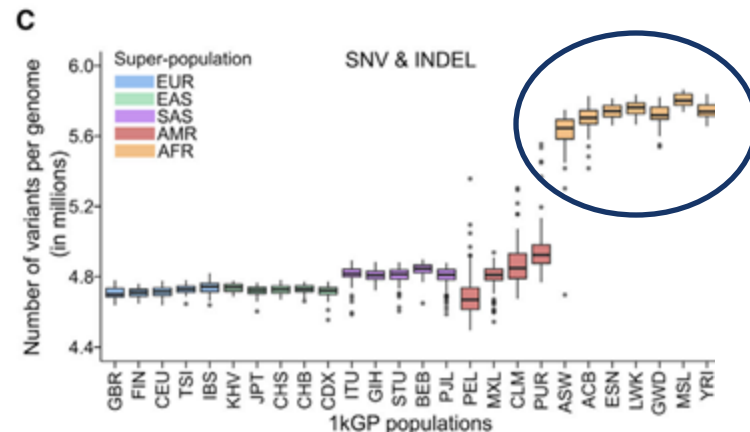
<https://towardsdatascience.com/how-exactly-umap-works-13e3040e1668>

1000 Genomes Project



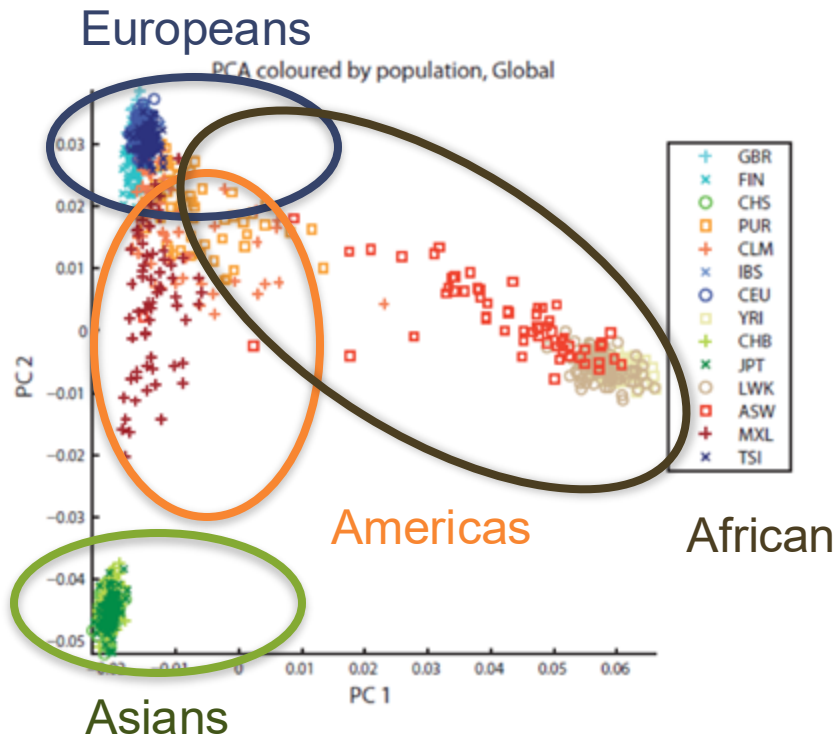
Why do people with African ancestry have so many more variants?

EAS	CHB	Han Chinese	Han Chinese in Beijing, China
	JPT	Japanese	Japanese in Tokyo, Japan
	CHS	Southern Han Chinese	Han Chinese South
	CDX	Dai Chinese	Chinese Dai in Xishuangbanna, China
	KHV	Kinh Vietnamese	Kinh in Ho Chi Minh City, Vietnam
EUR	CEU	Utah residents (CEPH) with Northern and Western European ancestry	Utah residents (CEPH) with Northern and Western European ancestry
	TSI	Tuscan	Tuscans in Italy
	GBR	British	British in England and Scotland
	FIN	Finnish	Finnish in Finland
	IBS	Spanish	Iberian populations in Spain
AFR	YRI	Yoruba	Yoruba in Ibadan, Nigeria
	LWK	Luhya	Luhya in Mbulue, Kenya
	GWD	Gambian	Gambian in Western Division, The Gambia
	MSL	Mende	Mende in Sierra Leone
	ESN	Esan	Esan in Nigeria
AMR	ASW	African-American SW	African Ancestry in Southwest US
	ACB	African-Caribbean	African Caribbean in Barbados
	MXL	Mexican-American	Mexican Ancestry in Los Angeles, California
	PUR	Puerto Rican	Puerto Rican in Puerto Rico
	CLM	Colombian	Colombian in Medellin, Colombia
SAS	PEL	Peruvian	Peruvian in Lima, Peru
	GJH	Gujarati	Gujarati Indian in Houston, TX
	PJL	Punjabi	Punjabi in Lahore, Pakistan
	BEB	Bengali	Bengali in Bangladesh
	STU	Sri Lankan	Sri Lankan Tamil in the UK
	ITU	Indian	Indian Telugu in the UK



High-coverage whole-genome sequencing of the expanded 1000 Genomes Project cohort including 602 trios
 Byrska-Bishop et al. (2022) Cell. doi: 10.1016/j.cell.2022.08.004

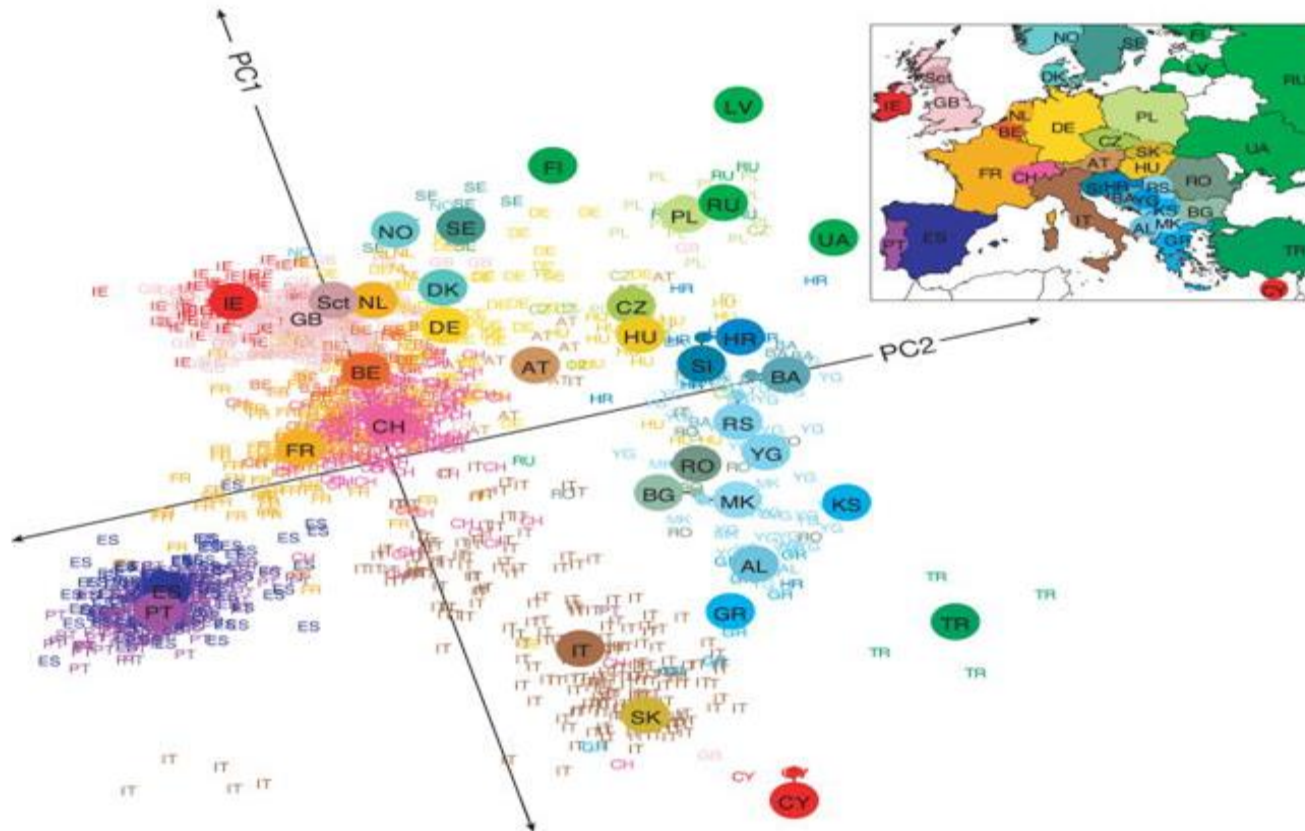
Variation across populations



	Person1	Person2	Person3	...
SNP1	0/0	0/1	0/1	
SNP2	0/1	0/1	0/1	
SNP3	0/0	0/0	1/1	
SNP4	1/1	0/1	1/1	
SNP5	0/0	1/1	1/1	
SNP5	0/0	0/0	0/1	
...				

PC1 is largely correlated with African ancestry:
Human genetic diversity is dominated by African ancestry

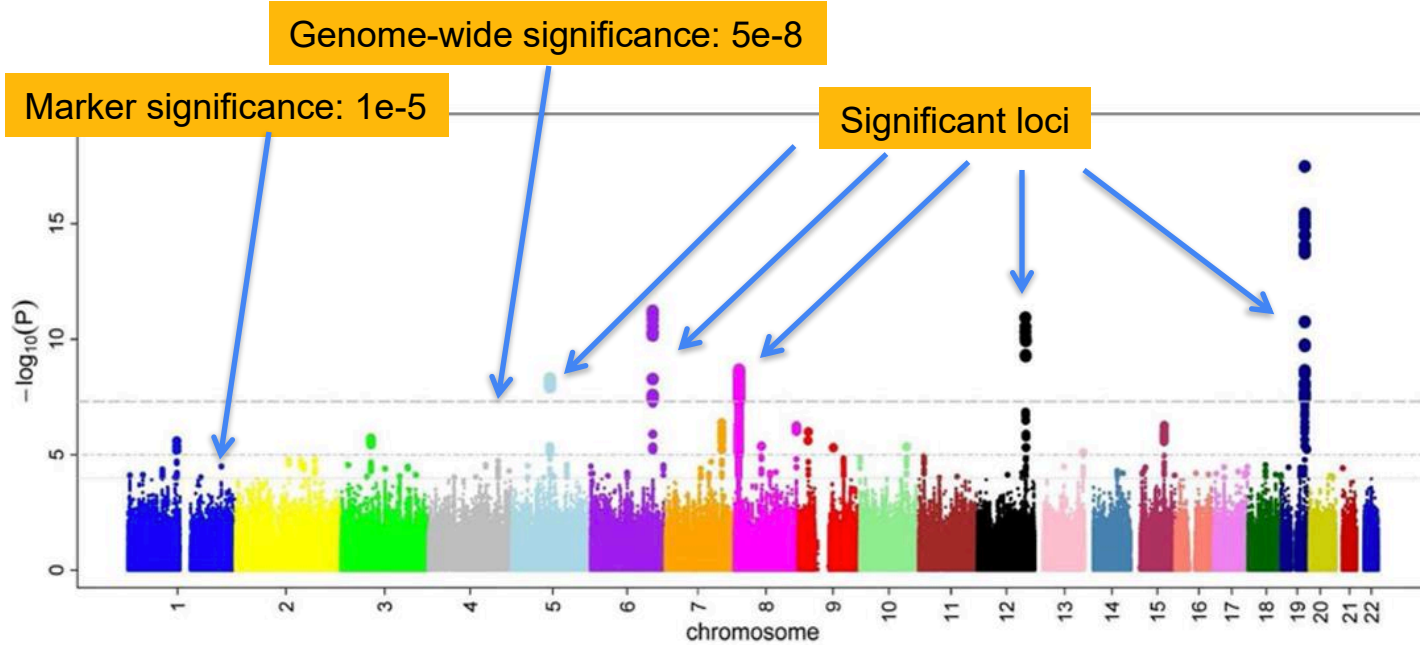
a



Genes mirror geography within Europe

Novembre et al (2008) Nature. doi: 10.1038/nature07331

Manhattan Plot



Four Novel Loci (19q13, 6q24, 12q24, and 5q14) Influence the Microcirculation In Vivo

Ikram et al (2010) PLOS Genetics. doi: 10.1371/journal.pgen.1001184

GWAS Catalog

As of 2025-09-15, the GWAS Catalog contains 7410 publications,
1,008,330 top associations and 139,254 full summary statistics.



AlexNet

“AlexNet”

ImageNet Classification with Deep Convolutional Neural Networks

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Abstract

We trained a large, deep convolutional neural network to classify the 1.2 million high-resolution images in the ImageNet ILSVRC-2010 contest into the 1000 different classes. On the test data, we achieved top-1 and top-5 error rates of 37.5% and 17.0% which is considerably better than the previous state-of-the-art. The neural network, which has 60 million parameters and 650,000 neurons, consists of five convolutional layers, some of which are followed by max-pooling layers, and three fully-connected layers with a final 1000-way softmax. To make training faster, we used non-saturating neurons and a very efficient GPU implementation of the convolution operation. To reduce overfitting in the fully-connected layers we employed a recently-developed regularization method called “dropout” that proved to be very effective. We also entered a variant of this model in the ILSVRC-2012 competition and achieved a winning top-5 test error rate of 15.3%, compared to 26.2% achieved by the second-best entry.



Alex Krizhevsky
U. Toronto/Google



Ilya Sutskever
U. Toronto/OpenAI/
Safe Superintelligence Inc



Geoffrey Hinton
U. Toronto/Vector Institute

“AlexNet”

Imagenet classification with deep convolutional neural networks

Authors Alex Krizhevsky, Ilya Sutskever, Geoffrey E Hinton

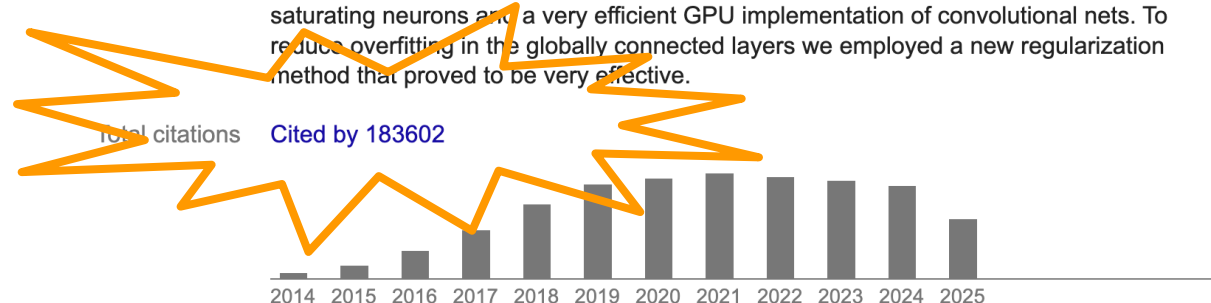
Publication date 2012

Journal Advances in neural information processing systems

Volume 25

Description We trained a large, deep convolutional neural network to classify the 1.3 million high-resolution images in the LSVRC-2010 ImageNet training set into the 1000 different classes. On the test data, we achieved top-1 and top-5 error rates of 39.7\% and 18.9\% which is considerably better than the previous state-of-the-art results. The neural network, which has 60 million parameters and 500,000 neurons, consists of five convolutional layers, some of which are followed by max-pooling layers, and two globally connected layers with a final 1000-way softmax. To make training faster, we used non-saturating neurons and a very efficient GPU implementation of convolutional nets. To reduce overfitting in the globally connected layers we employed a new regularization method that proved to be very effective.

Total citations Cited by 183602



Year	Total citations
2014	100
2015	200
2016	400
2017	600
2018	800
2019	1000
2020	1200
2021	1400
2022	1400
2023	1300
2024	1200
2025	1000

Outline

1. The Problem
2. pre-AlexNet approaches
3. How do animals & humans approach this?
4. Artificial Neural Networks
5. Convolutional Neural Networks
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What is this?



What is this?



Fish



Axe



Bagel



Dog



Van

ImageNet Classification Challenge

SUN, 131K

[Xiao et al. '10]

LabelMe, 37K

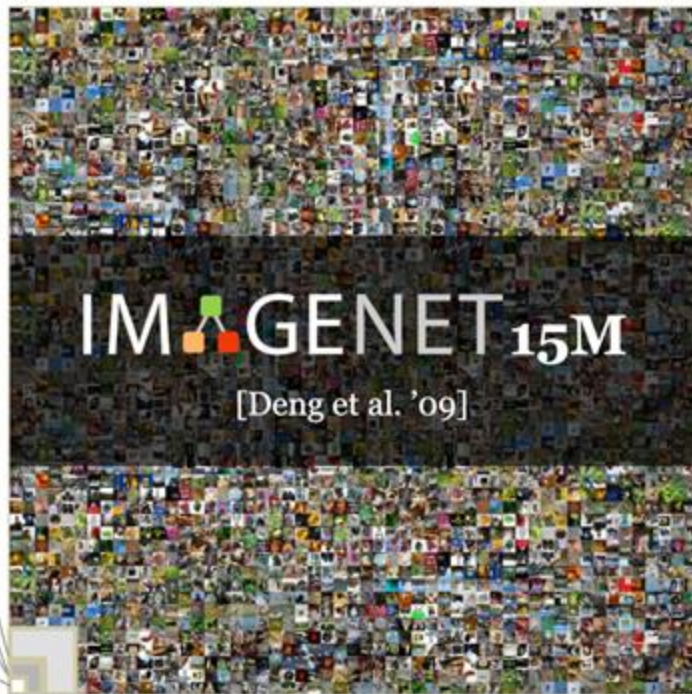
[Russell et al. '07]

PASCAL VOC, 30K

[Everingham et al. '06-'12]

Caltech101, 9K

[Fei-Fei, Fergus, Perona, '03]



- 15M images organized into 21,841 categories
- ILSVRC (AlexNet) uses a subset:
 - 1000 images in 1000 categories
 - 1.2M training, 50k validation, 150k test

imagenet_ilsvrc2017_v1.0

ImageNet Examples

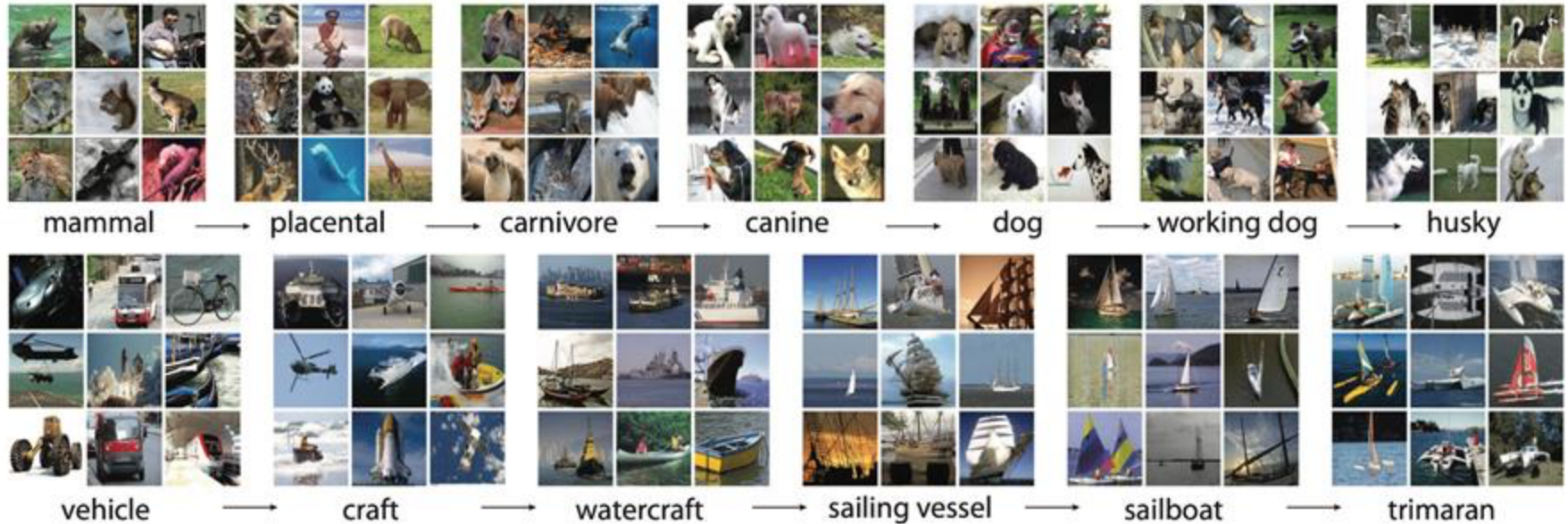


Figure 1: A snapshot of two root-to-leaf branches of ImageNet: the **top** row is from the mammal subtree; the **bottom** row is from the vehicle subtree. For each synset, 9 randomly sampled images are presented.

Images organized into nested categories determined by “WordNet”

Classification Challenges



husky

- “Natural” variation between individuals
 - Different colors, shapes, sizes
 - Very similar categories: 90 different dog breeds in the challenge (120 total)
- “Image” variations
 - Orientations: side, front, behind, etc
 - Distance & position of the object
 - Other objects in the image

Classification Challenges



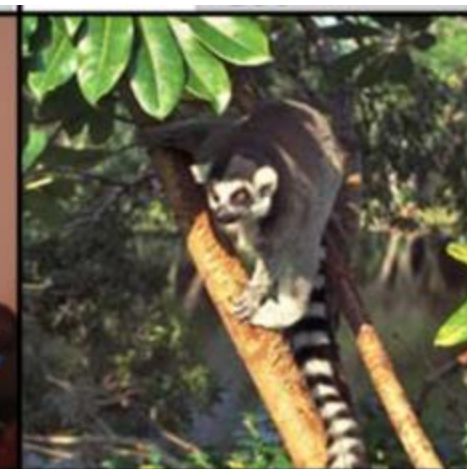
grille



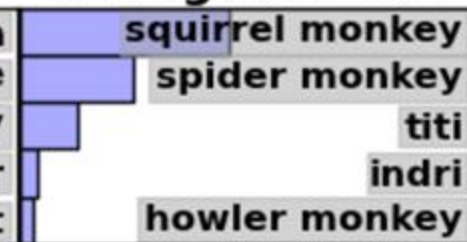
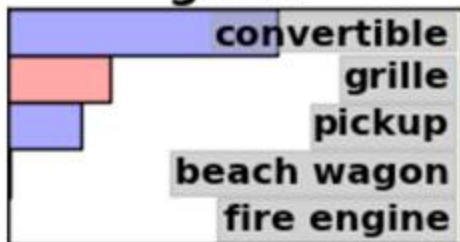
mushroom



cherry



Madagascar cat



Classification “errors” from AlexNet

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Abstract

We address image classification on a large-scale, i.e. when a large number of images and classes are involved. First, we study classification accuracy as a function of the image signature dimensionality and the training set size. We show experimentally that the larger the training set, the higher the impact of the dimensionality on the accuracy. In other words, high-dimensional signatures are important to obtain state-of-the-art results on large datasets. Second, we tackle the problem of data compression on very large signatures (on the order of 10^5 dimensions) using two lossy compression strategies: a dimensionality reduction technique known as the hash kernel and an encoding technique based on product quantizers. We explain how the gain in storage can be traded against a loss in accuracy and/or an increase in CPU cost. We report results on two large databases – ImageNet and a dataset of 1M Flickr images – showing that we can reduce the storage of our signatures by a factor 64 to 128 with little loss in accuracy. Integrating the decompression in the classifier learning yields an efficient and scalable training algorithm. On ILSVRC2010 we report a 74.3% accuracy at top-5, which corresponds to a 2.5% absolute improvement with respect to the state-of-the-art. On a subset of 10K classes of ImageNet we report a top-1 accuracy of 16.7%, a relative improvement of 160% with respect to the state-of-the-art.

winners of the PASCAL VOC 2007 [8] and 2008 [9] competitions used a similar paradigm: many types of low-level local features are extracted (referred to as “channels”), one bag-of-visual-words (BOV) histogram is computed for each channel and non-linear kernel classifiers such as SVMs are used to perform classification [38, 29]. The use of many channels and costly non-linear SVMs was made possible by the modest size of the available databases.

In recent years only has the computational cost become a central issue in image classification / object detection. In [19], Maji *et al.* showed that the runtime cost of an intersection kernel (IK) SVM could be made independent of the number of support vectors. Maji and Berg [18] and Wang *et al.* [31] then proposed efficient algorithms to learn IK-SVMs. Vedaldi and Zisserman [30] and Perronnin *et al.* [21] subsequently generalized this principle to any additive classifier. Another line of research consists in computing image representations which are directly amenable to cost-less linear classification. Yang *et al.* [36], Wang *et al.* [32] and Boureau *et al.* [4] showed that replacing the average pooling stage in the BOV computation by a max-pooling yielded excellent results. To go beyond the BOV, i.e. beyond counting, it has been proposed to include higher order statistics in the image signature. This includes modeling an image by a probability distribution [17, 35] or using the Fisher kernel framework [20]. Especially, it was shown that the Fisher Vector (FV) could yield high accuracy with linear classifiers [22].

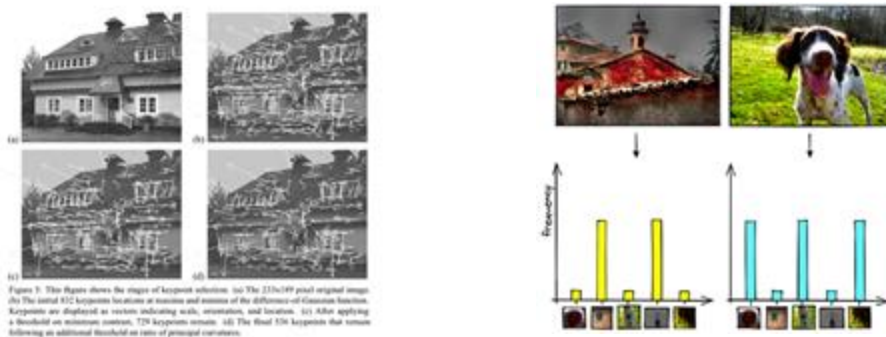
If one wants to stick to efficient linear classifiers, the image representations should be high-dimensional to ensure linear separability of the classes. Therefore, we argue that the storage/memory cost is becoming a central issue in large-scale image classification. As an example, in this paper we consider almost dense image representations – based on the improved FV of [22] – with up to 524K dimensions. Using a 4 byte floating point representation, a single signature requires 2MB of storage. Storing the ILSVRC2010 dataset [2] would take approximately 2.8TBs and storing the full ImageNet dataset around 23TBs. Obviously, these numbers have to be multiplied by the number of channels, i.e. feature types. As another example, the

1. Break an image into (many) patches
2. Transform local features into “visual words” using a “code book”
3. Use the visual words to create a sparse feature vector
4. Compare the vector to the vectors precomputed from your database

1. Introduction

Scaling-up image classification systems is a problem which is receiving an increasing attention as larger labeled image datasets are becoming available. For instance, ImageNet (www.image-net.org) consists of more than 12M images of 17K concepts [7] and Flickr contains thousands of groups (www.flickr.com/groups/) – some of which with hundreds of thousands of pictures – which can be readily used to learn object classifiers [31, 22].

The focus in the image classification community was initially on developing systems which would yield the best possible accuracy fairly independently of their cost. The



The winner in 2011

“SIFT descriptors”

“Visual word histogram”

Jorge Sánchez and Florent Perronnin
Textual and Visual Pattern Analysis (TVPA) group
Xerox Research Centre Europe (XRCE)

Abstract

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winner of the PASCAL VOC 2007 [8] and 2008 [9] competitions used a similar paradigm: many types of low-level local features are extracted (referred to as “channels”), one bag-of-visual-words (BOV) histogram is computed for each channel and non-linear kernel classifiers such as SVMs are used to perform classification [38, 29]. The use of many channels and costly non-linear SVMs was made possible by the modest size of the available databases.

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If one wants to stick to efficient linear classifiers, the image representations should be high-dimensional to ensure linear separability of the classes. Therefore, we argue that *the storage/memory cost is becoming a central issue in large-scale image classification*. As an example, in this paper we consider almost dense image representations – based on the improved FV of [22] – with up to 524K dimensions. Using a 4 byte floating point representation, a single signature requires 2MB of storage. Storing the ILSVRC2010 dataset [2] would take approximately 2.8TBs and storing the full ImageNet dataset around 23TBs. Obviously, these numbers have to be multiplied by the number of channels, i.e. feature types. As another example, the

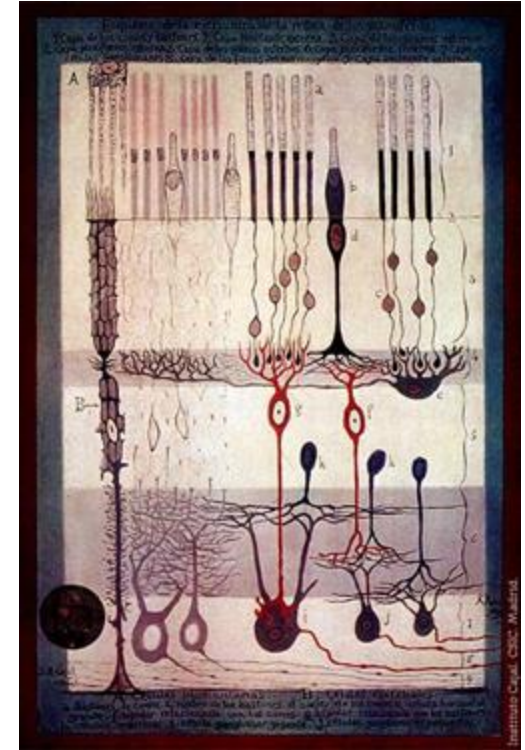
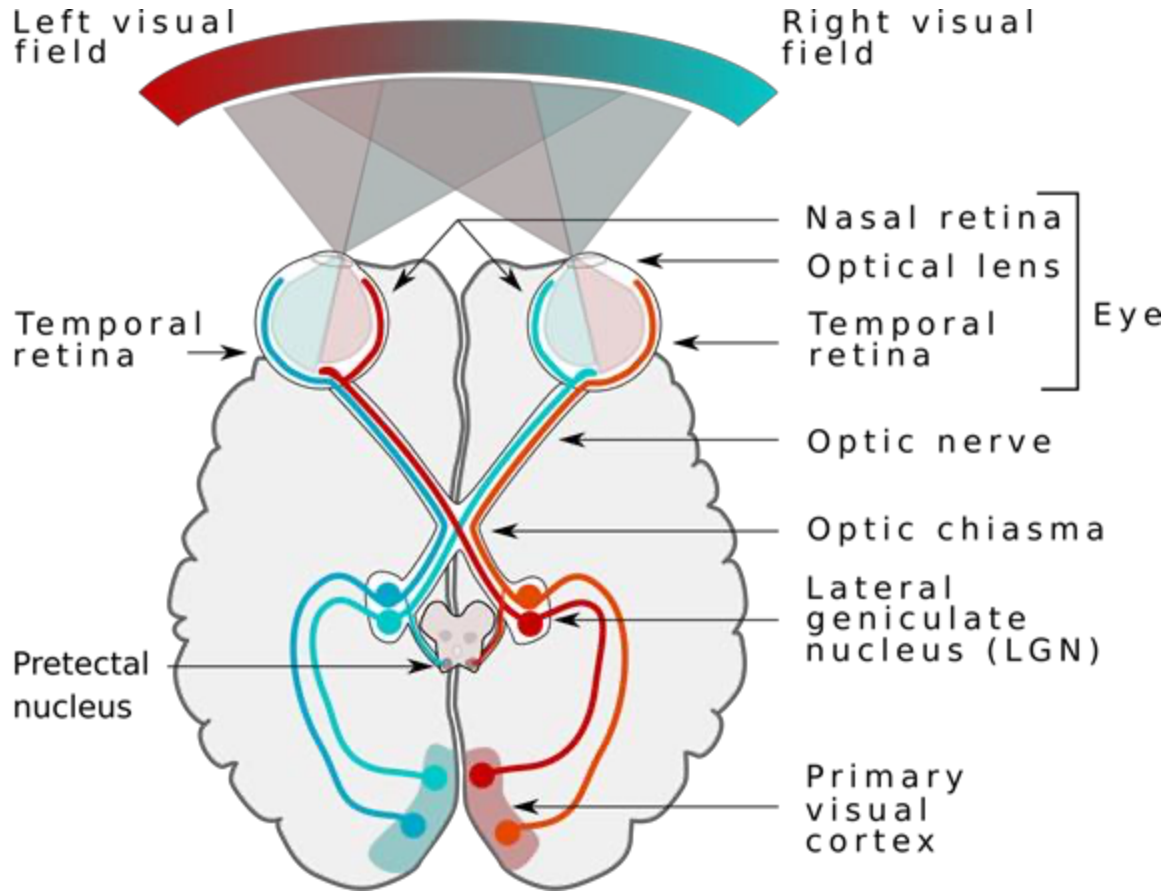
Limitations

- Ignores spatial relationships between patches (“a bag of words”)
- High dependency of local features that are (often) hand crafted to maximize performance on the training data
- Computationally demanding to extract features, perform high dimensional comparisons
- >25% error

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Human Visual Cortex



Structure of the Mammalian Retina
S. Ramón y Cajal (1900)

Vision & Pattern Recognition

Receptive Fields, Binocular Interaction and Functional Architecture in the Cat's Visual Cortex

D. H. HUBEL AND T. N. WIESEL • Neurophysiology Laboratory, Department of Pharmacology Harvard Medical School, Boston, Massachusetts

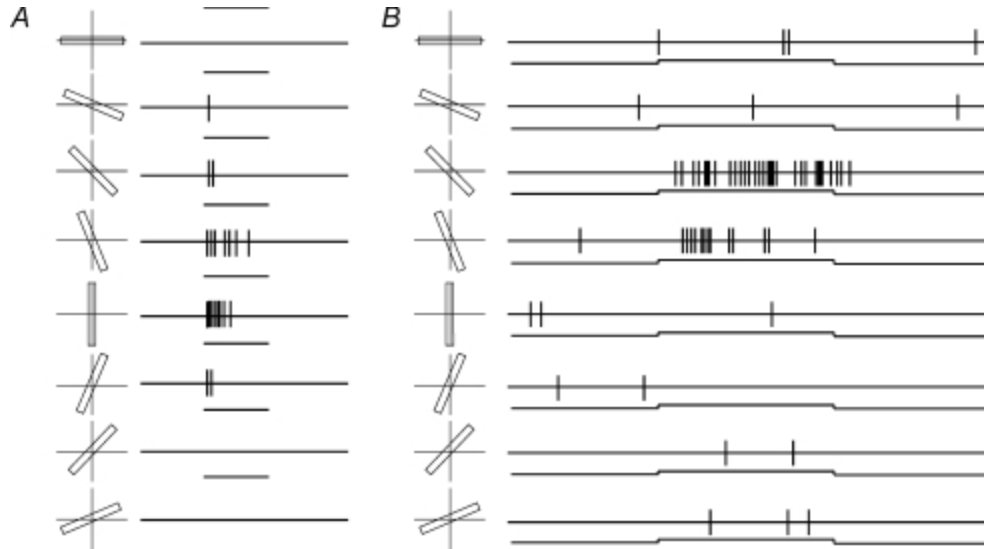


Table 1. Simple Cortical Fields

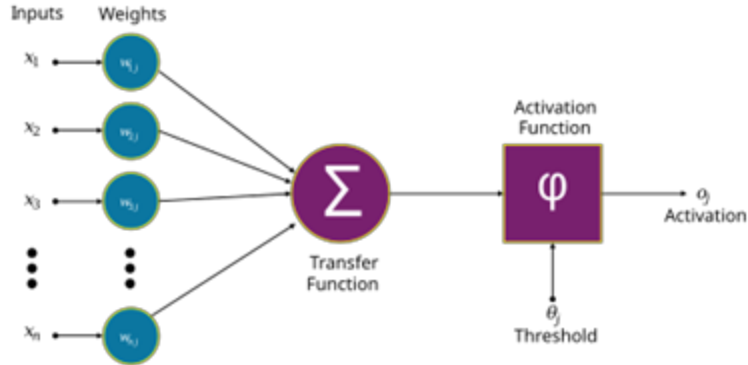
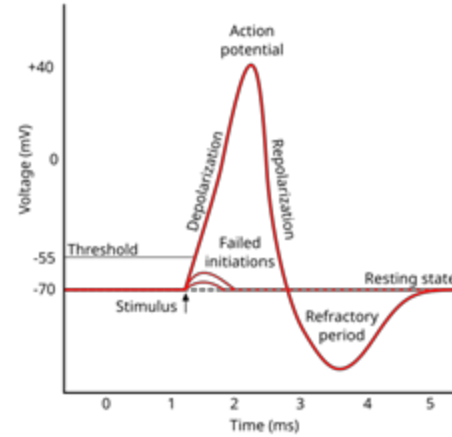
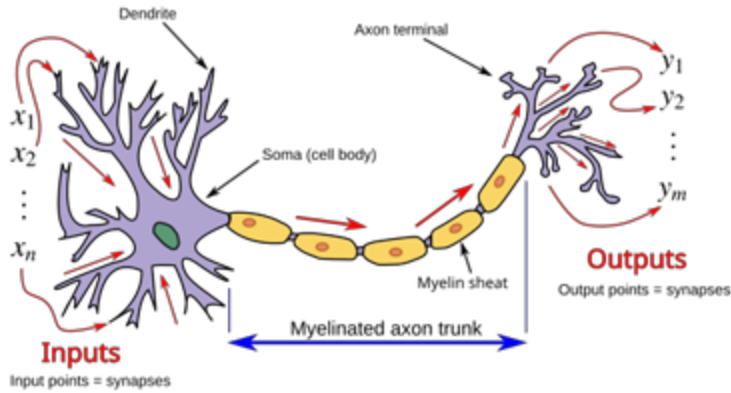
	Text-fig.	No. of cells
(a) Narrow concentrated centres		
(i) Symmetrical flanks		
Excitatory centres	2C	23
Inhibitory centres	2D	17
(ii) Asymmetrical flanks		
Excitatory centres	—	28
Inhibitory centres	2E	10
(b) Large centres; concentrated flanks	2F	21
(c) One excitatory region and one inhibitory	2G	17
(d) Uncategorized	—	117
Total number of simple fields		233

Example simple cells in the striate cortex of anaesthetized paralysed cat (A) and the awake fixating monkey (B). The comparison illustrates the similar response to oriented slits of light in the anaesthetized, paralysed cats and awake behaving monkeys. The example from the awake monkey shows the same qualitative orientation tuning but a slightly higher background rate than from the anaesthetized cat. Traced from Fig. 3A of Hubel & Wiesel (1959) and from Fig. 5 of (Wurtz, 1969c).

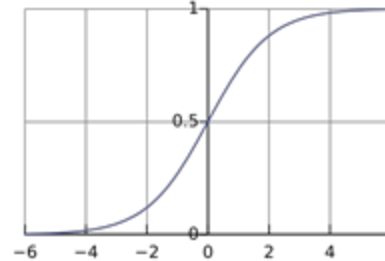
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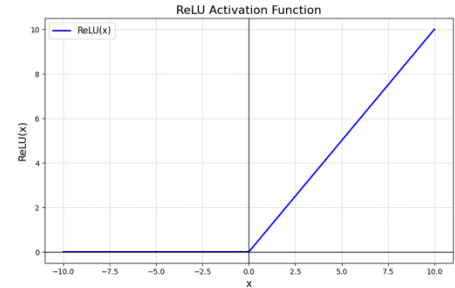
Biological vs Artificial Neurons



sigmoid

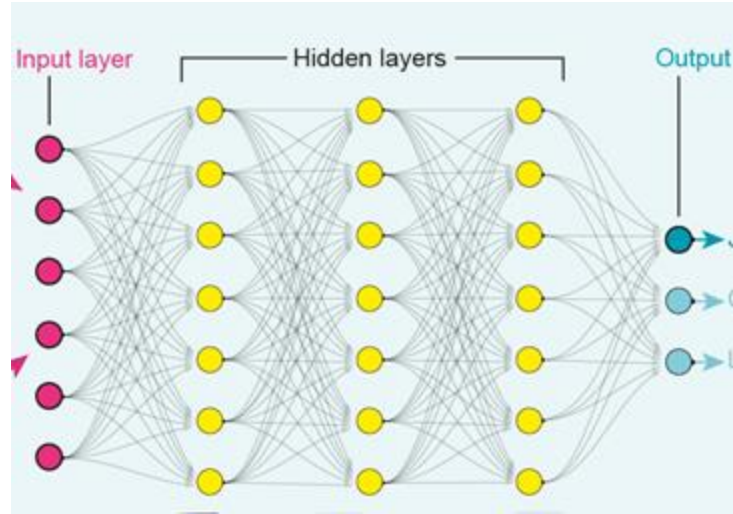


relu

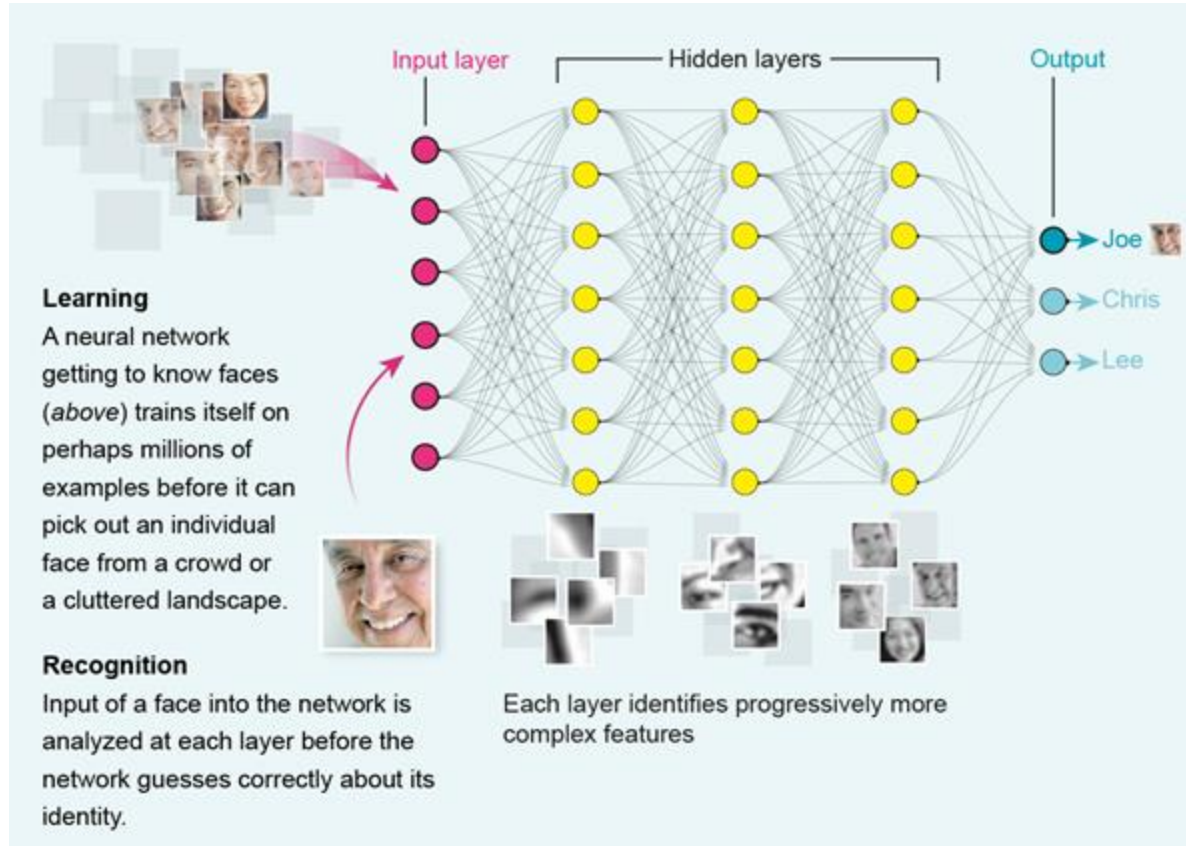


The non-linear activation function (sigmoid/relu) is needed to model non-linear relationships

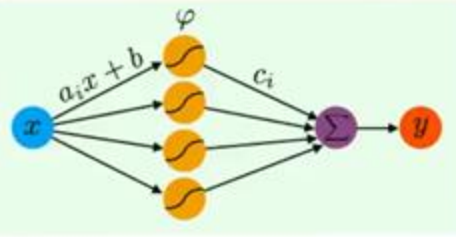
Artificial Neural Networks



Artificial Neural Networks



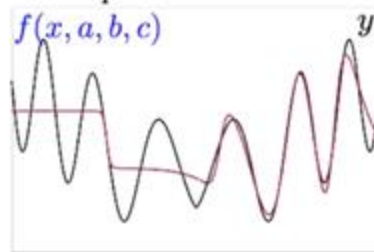
ANNs are “Universal Approximators”



1 hidden layer perceptron:

$$y \approx f(x, a, b, c) \stackrel{\text{def.}}{=} \sum_{i=1}^p c_i \varphi(a_i x + b_i)$$

$p = 6$ neurons



$p = 20$ neurons



Approximation by Superpositions of a Sigmoidal Function*

G. Cybenko†

Abstract. In this paper we demonstrate that finite linear combinations of compositions of a fixed, univariate function and a set of affine functionals can uniformly approximate any continuous function of n real variables with support in the unit hypercube; only mild conditions are imposed on the univariate function. Our results settle an open question about representability in the class of single hidden layer neural networks. In particular, we show that arbitrary decision regions can be arbitrarily well approximated by continuous feedforward neural networks with only a single internal, hidden layer and any continuous sigmoidal nonlinearity. The paper discusses approximation properties of other possible types of nonlinearities that might be implemented by artificial neural networks.

Key words. Neural networks, Approximation, Completeness.



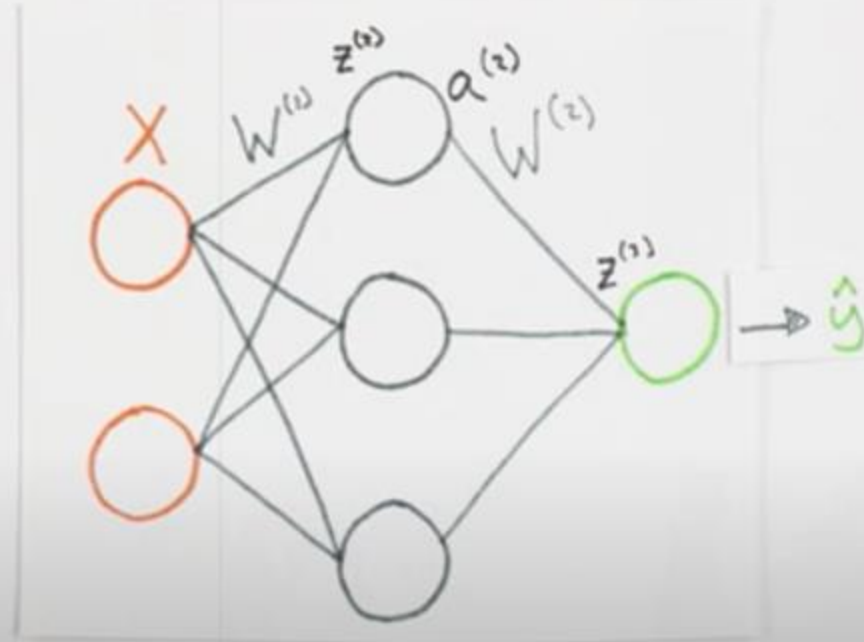
Training with Backpropagation: Random initialization

```
class Neural_Network(object):
    def __init__(self):
        #Define Hyperparameters
        self.inputLayerSize = 2
        self.outputLayerSize = 1
        self.hiddenLayerSize = 3

        #Weights (Parameters)
        self.W1 = np.random.randn(self.inputLayerSize, \
                                    self.hiddenLayerSize)
        self.W2 = np.random.randn(self.hiddenLayerSize, \
                                    self.outputLayerSize)

    def forward(self, X):
        #Propagate inputs though network
        self.z2 = np.dot(X, self.W1)
        self.a2 = self.sigmoid(self.z2)
        self.z3 = np.dot(self.a2, self.W2)
        yHat = self.sigmoid(self.z3)
        return yHat

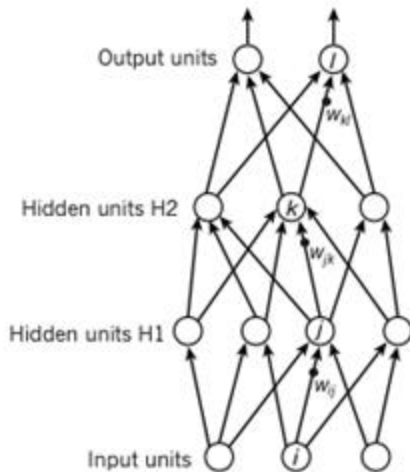
    def sigmoid(self, z):
        #Apply sigmoid activation function to scalar, vector, or matrix
        return 1/(1+np.exp(-z))
```



Training with Backpropagation

Forward pass: evaluation

c



$$y_l = f(z_l) \\ z_l = \sum_{k \in H2} w_{kl} y_k$$

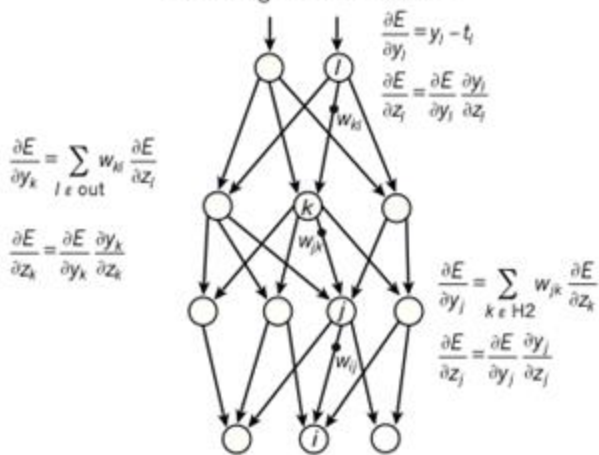
$$y_k = f(z_k) \\ z_k = \sum_{j \in H1} w_{jk} y_j$$

$$y_j = f(z_j) \\ z_j = \sum_{i \in \text{Input}} w_{ij} x_i$$

Reverse pass: backpropagation

d

Compare outputs with correct answer to get error derivatives



$$\frac{\partial E}{\partial y_k} = \sum_{l \in \text{out}} w_{kl} \frac{\partial E}{\partial z_l}$$

$$\frac{\partial E}{\partial z_k} = \frac{\partial E}{\partial y_k} \frac{\partial y_k}{\partial z_k}$$

$$\frac{\partial E}{\partial y_l} = y_l - t_l \\ \frac{\partial E}{\partial z_l} = \frac{\partial E}{\partial y_l} \frac{\partial y_l}{\partial z_l}$$

$$\frac{\partial E}{\partial y_j} = \sum_{k \in H2} w_{jk} \frac{\partial E}{\partial z_k} \\ \frac{\partial E}{\partial z_j} = \frac{\partial E}{\partial y_j} \frac{\partial y_j}{\partial z_j}$$

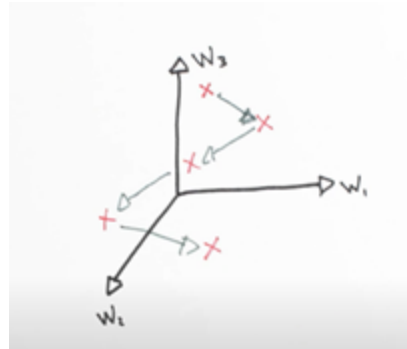
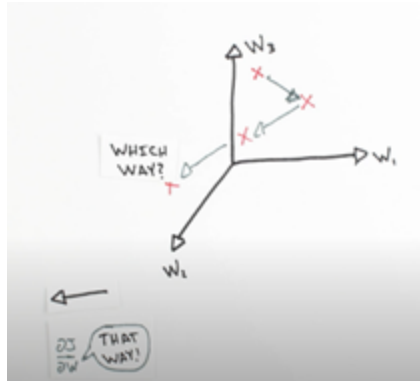
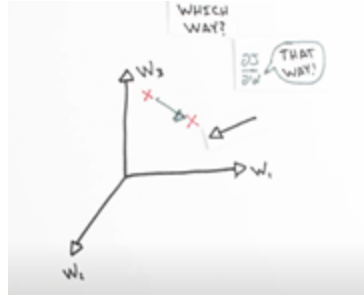
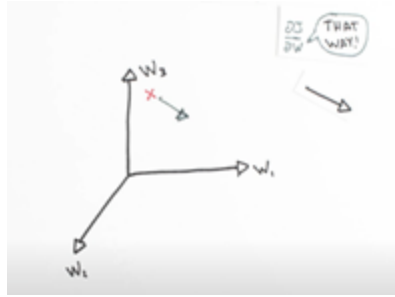
- Calculations along every edge, but the math is generally very basic
 - Addition, multiplication, exponentiation, etc
- Also easy to parallelize (especially with GPUs)

Deep Learning

LeCun, Bengio, Hinton (2020). Nature. doi:10.1038/nature14539

Training with Backpropagation: Gradient Descent

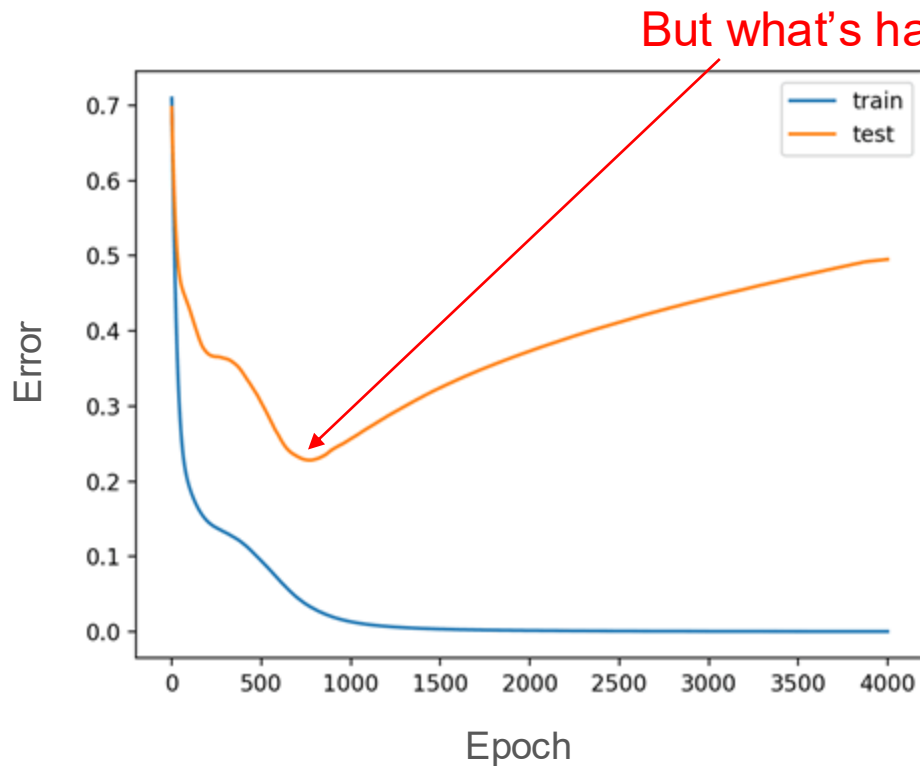
Minimize errors between predictions and true labels



Ball rolling down a hill: start with a random initialization, then stochastic gradient descent by iteratively computing the partial derivatives of the functions

Training ANN through multiple epochs

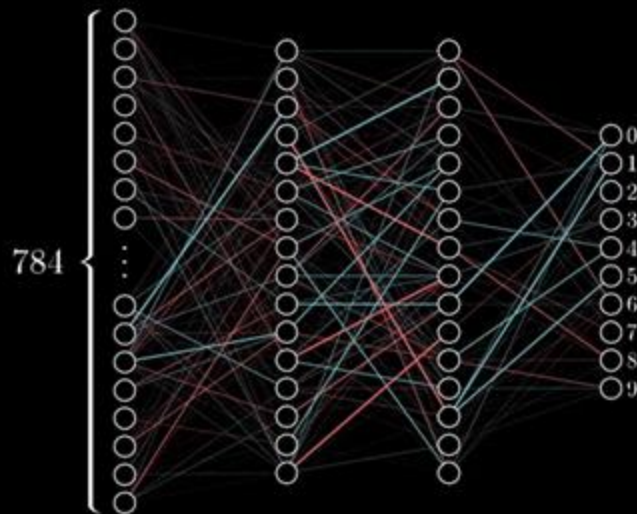
Error goes down with more training epochs



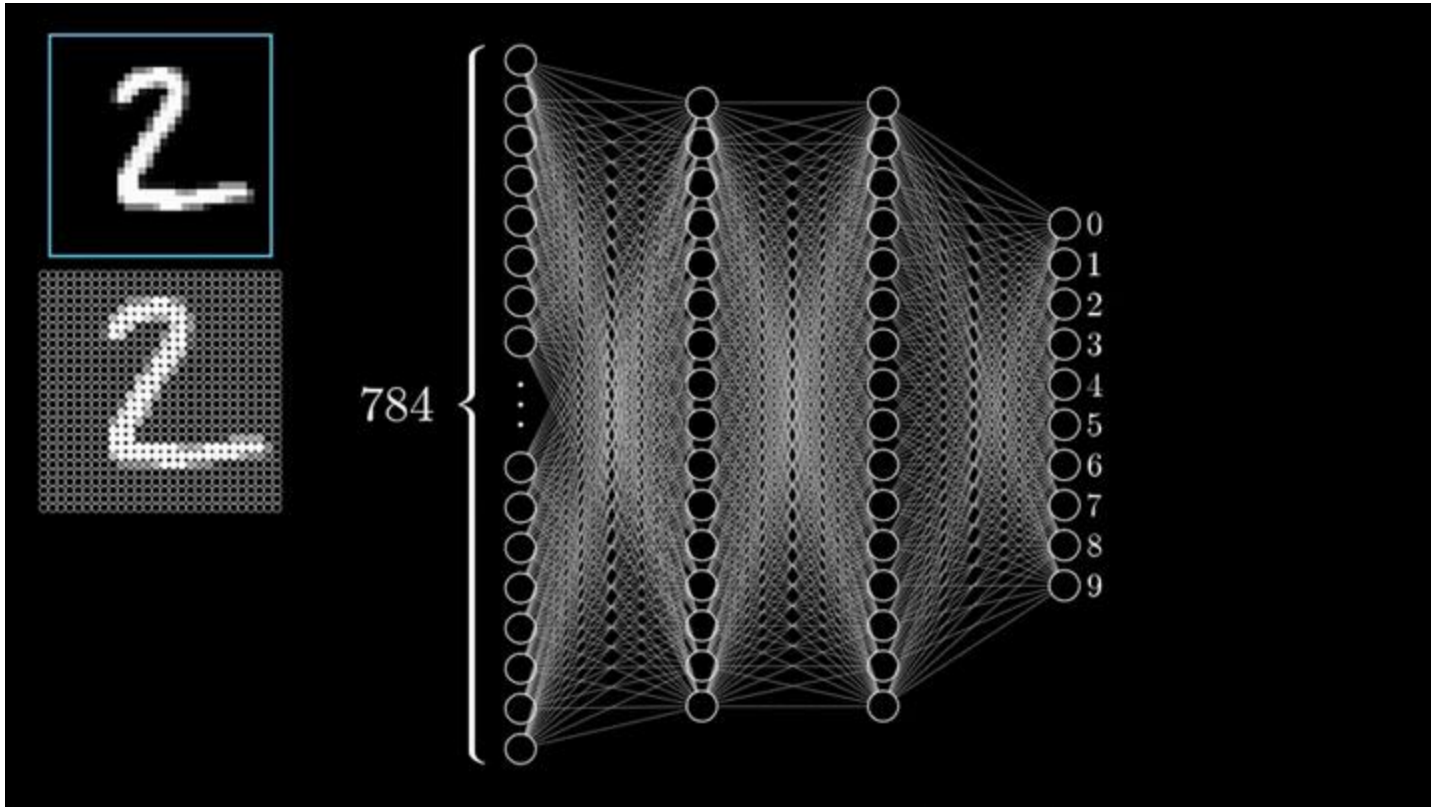
Training for too long
can lead to
overfitting!

Recap - Training

Training in
progress...



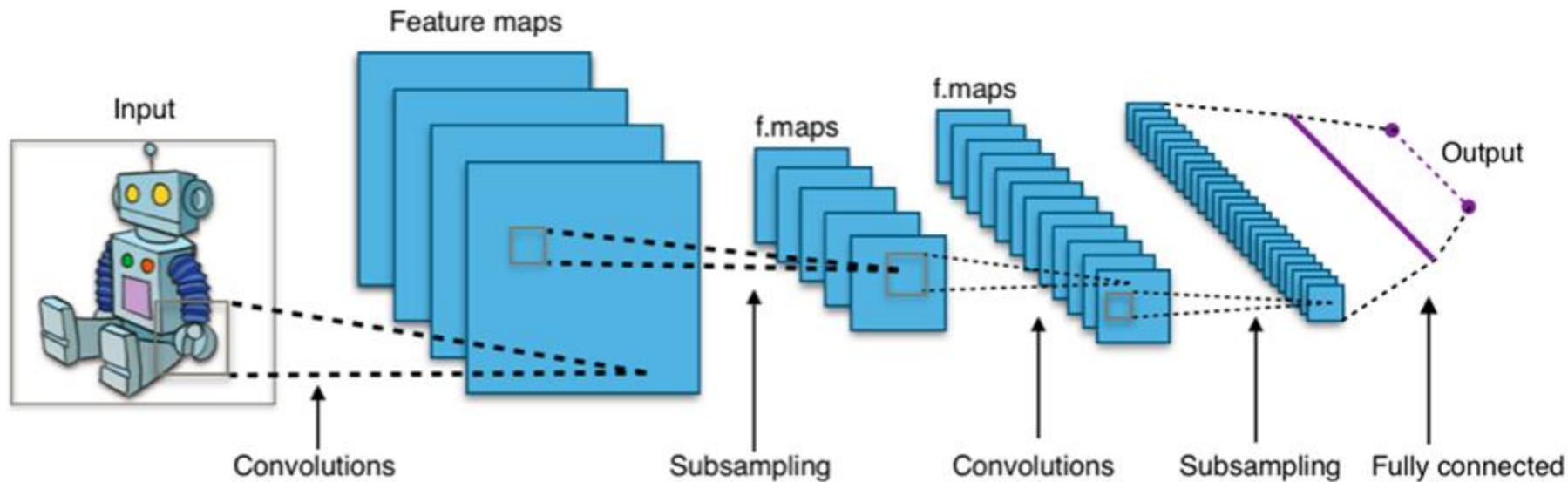
Recap - Evaluation



Outline

1. The Problem
2. pre-AlexNet approaches
3. How do animals & humans approach this?
4. Artificial Neural Networks
5. Convolutional Neural Networks
6. AlexNet
7. Impact

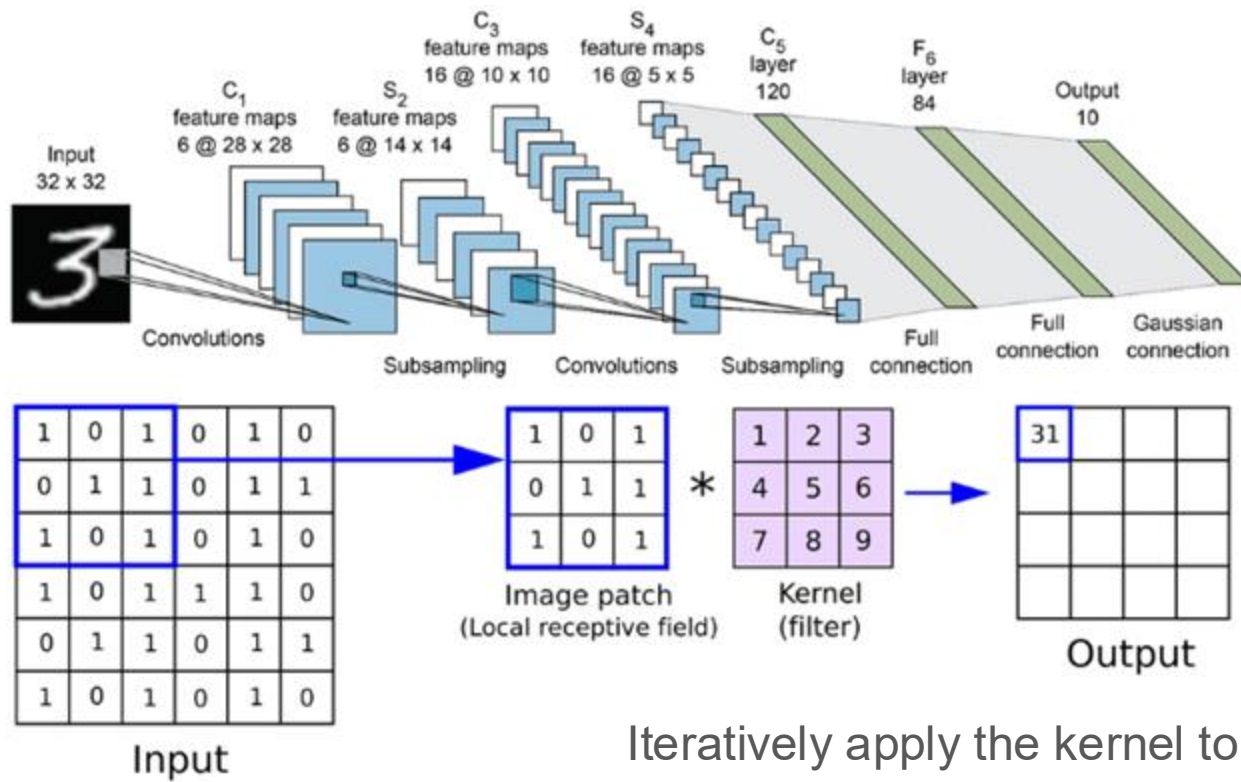
Convolutional Neural Network



Fully connected neural networks are most expressive but are computationally intensive!
CNNs achieve nearly the same power but with much less compute:

- $O(\text{Nodes} * \text{Depth})$ rather than $O(\text{Nodes} ^ \text{Depth})$

Convolutional Neural Network



“LeNet” (Lecun et al, 1998)

Iteratively apply the kernel to extract features

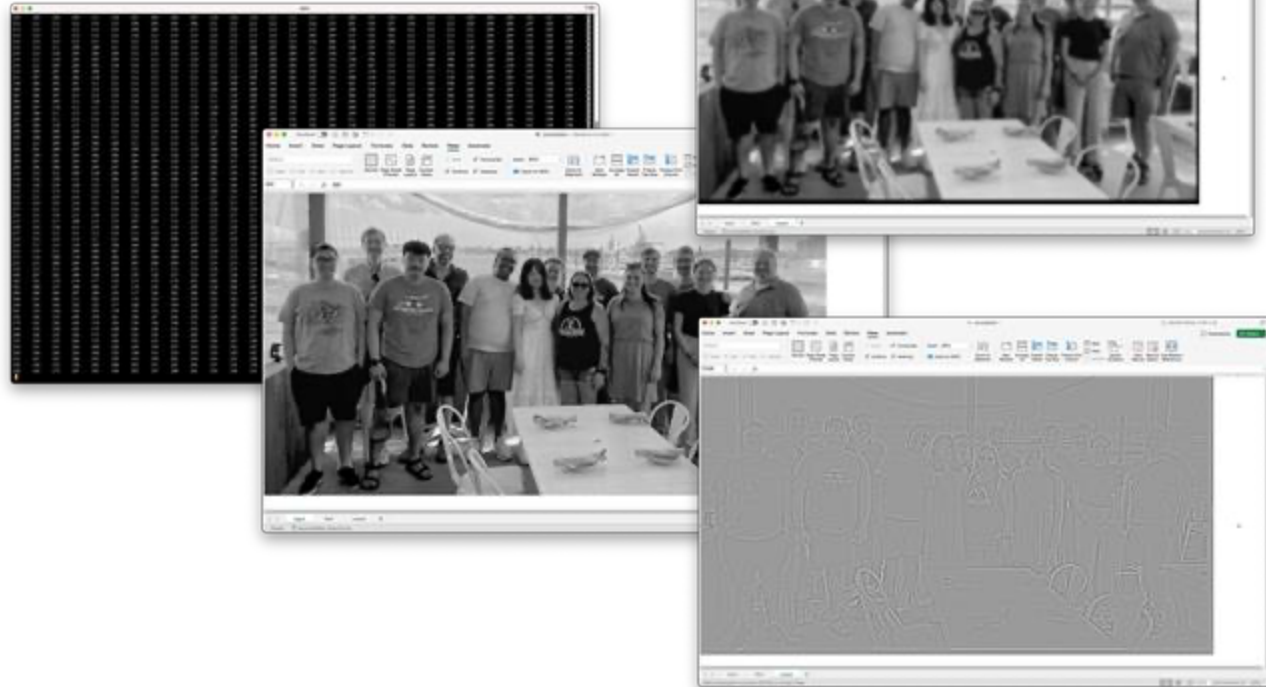
- Starts from the input image, then creates more abstract derived images

Kernels

Operation	Kernel u	Image result $g(x,y)$
Identity	$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$	
Ridge or edge detection	$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix}$	
	$\begin{bmatrix} -1 & -1 & -1 \\ -1 & 8 & -1 \\ -1 & -1 & -1 \end{bmatrix}$	
Sharpen	$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 5 & -1 \\ 0 & -1 & 0 \end{bmatrix}$	
Box blur (normalized)	$\frac{1}{9} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$	
Gaussian blur 3×3 (approximation)	$\frac{1}{16} \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$	
Gaussian blur 5×5 (approximation)	$\frac{1}{256} \begin{bmatrix} 1 & 4 & 6 & 4 & 1 \\ 4 & 16 & 24 & 16 & 4 \\ 6 & 24 & 36 & 24 & 6 \\ 4 & 16 & 24 & 16 & 4 \\ 1 & 4 & 6 & 4 & 1 \end{bmatrix}$	
Unsharp masking 5×5 Based on Gaussian blur with amount as 1 and threshold as 0 (with no image mask)	$\frac{-1}{256} \begin{bmatrix} 1 & 4 & 6 & 4 & 1 \\ 4 & 16 & 24 & 16 & 4 \\ 6 & 24 & -476 & 24 & 6 \\ 4 & 16 & 24 & 16 & 4 \\ 1 & 4 & 6 & 4 & 1 \end{bmatrix}$	

Kernels

Operation	Kernel u	Image result $g(x,y)$
Identity	$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$	
Ridge or edge detection	$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix}$	
	$\begin{bmatrix} -1 & -1 & -1 \\ -1 & 8 & -1 \\ -1 & -1 & -1 \end{bmatrix}$	
Sharpen	$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 5 & -1 \\ 0 & -1 & 0 \end{bmatrix}$	
Box blur (normalized)	$\frac{1}{9} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$	
Gaussian blur 3 x 3 (approximation)	$\frac{1}{16} \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$	
Gaussian blur 5 x 5 (approximation)	$\frac{1}{256} \begin{bmatrix} 1 & 4 & 6 & 4 & 1 \\ 4 & 16 & 24 & 16 & 4 \\ 6 & 24 & 36 & 24 & 6 \\ 4 & 16 & 24 & 16 & 4 \\ 1 & 4 & 6 & 4 & 1 \end{bmatrix}$	
Unsharp masking 5 x 5 Based on Gaussian blur with amount as 1 and threshold as 0 (with no image mask)	$\frac{-1}{256} \begin{bmatrix} 1 & 4 & 6 & 4 & 1 \\ 4 & 16 & 24 & 16 & 4 \\ 6 & 24 & 36 & 24 & 6 \\ 4 & 16 & 24 & 16 & 4 \\ 1 & 4 & 6 & 4 & 1 \end{bmatrix}$	



$$= \text{input!A1} * \text{filter!\$A\$1} + \text{input!A2} * \text{filter!\$A\$2} + \text{input!A3} * \text{filter!\$A\$3} + \text{input!A4} * \text{filter!\$A\$4} + \text{input!A5} * \text{filter!\$A\$5} +$$

$$\text{input!B1} * \text{filter!\$B\$1} + \text{input!B2} * \text{filter!\$B\$2} + \text{input!B3} * \text{filter!\$B\$3} + \text{input!B4} * \text{filter!\$B\$4} + \text{input!B5} * \text{filter!\$B\$5} +$$

$$\text{input!C1} * \text{filter!\$C\$1} + \text{input!C2} * \text{filter!\$C\$2} + \text{input!C3} * \text{filter!\$C\$3} + \text{input!C4} * \text{filter!\$C\$4} + \text{input!C5} * \text{filter!\$C\$5} +$$

$$\text{input!D1} * \text{filter!\$D\$1} + \text{input!D2} * \text{filter!\$D\$2} + \text{input!D3} * \text{filter!\$D\$3} + \text{input!D4} * \text{filter!\$D\$4} + \text{input!D5} * \text{filter!\$D\$5} +$$

$$\text{input!E1} * \text{filter!\$E\$1} + \text{input!E2} * \text{filter!\$E\$2} + \text{input!E3} * \text{filter!\$E\$3} + \text{input!E4} * \text{filter!\$E\$4} + \text{input!E5} * \text{filter!\$E\$5}$$

Outline

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7. Impact

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the authors' study may be generalized to other populations, especially if a confidence interval for the difference in the two groups is constructed. The authors' study may be generalized to other populations, especially if a confidence interval for the difference in the two groups is constructed. The authors' study may be generalized to other populations, especially if a confidence interval for the difference in the two groups is constructed.

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These approaches to the development of a curriculum require a number of conditions to be met. First, the curriculum must be based on a clear understanding of the needs of the community. This requires a thorough analysis of the community's current and future needs, as well as a clear understanding of the community's values and beliefs. Second, the curriculum must be based on a clear understanding of the needs of the individual. This requires a thorough analysis of the individual's current and future needs, as well as a clear understanding of the individual's values and beliefs. Third, the curriculum must be based on a clear understanding of the needs of the society. This requires a thorough analysis of the society's current and future needs, as well as a clear understanding of the society's values and beliefs. Fourth, the curriculum must be based on a clear understanding of the needs of the world. This requires a thorough analysis of the world's current and future needs, as well as a clear understanding of the world's values and beliefs. Finally, the curriculum must be based on a clear understanding of the needs of the future. This requires a thorough analysis of the future's current and future needs, as well as a clear understanding of the future's values and beliefs.

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It is also important to note that the authors of the study found that the use of a single, standardized measure of social support was not sufficient to capture the complexity of social support in the workplace. They argue that future research should focus on developing more comprehensive measures of social support that take into account the various ways in which social support can be provided in the workplace.

3. The Market

Figure 1. A schematic diagram of the 12-step process for developing a strategic plan. The process is divided into three main phases: **Phase 1: Assessment** (Steps 1-4), **Phase 2: Strategy Development** (Steps 5-8), and **Phase 3: Implementation and Evaluation** (Steps 9-12). The steps are interconnected, showing a sequential flow with feedback loops. Key components include: **Step 1: Vision Statement**, **Step 2: Mission Statement**, **Step 3: SWOT Analysis**, **Step 4: Strategic Objectives**, **Step 5: Strategic Initiatives**, **Step 6: Action Plans**, **Step 7: Resource Allocation**, **Step 8: Monitoring and Evaluation**, **Step 9: Implementation**, **Step 10: Reporting**, **Step 11: Review**, and **Step 12: Revision**.

22

3 The architecture

Network Architecture

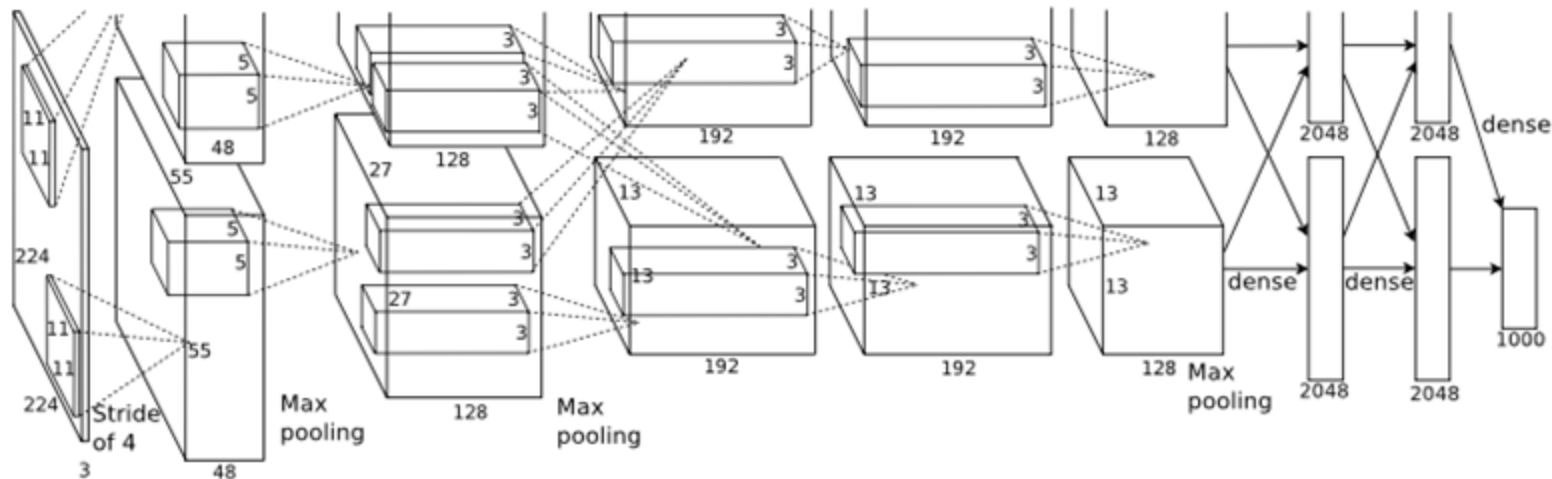
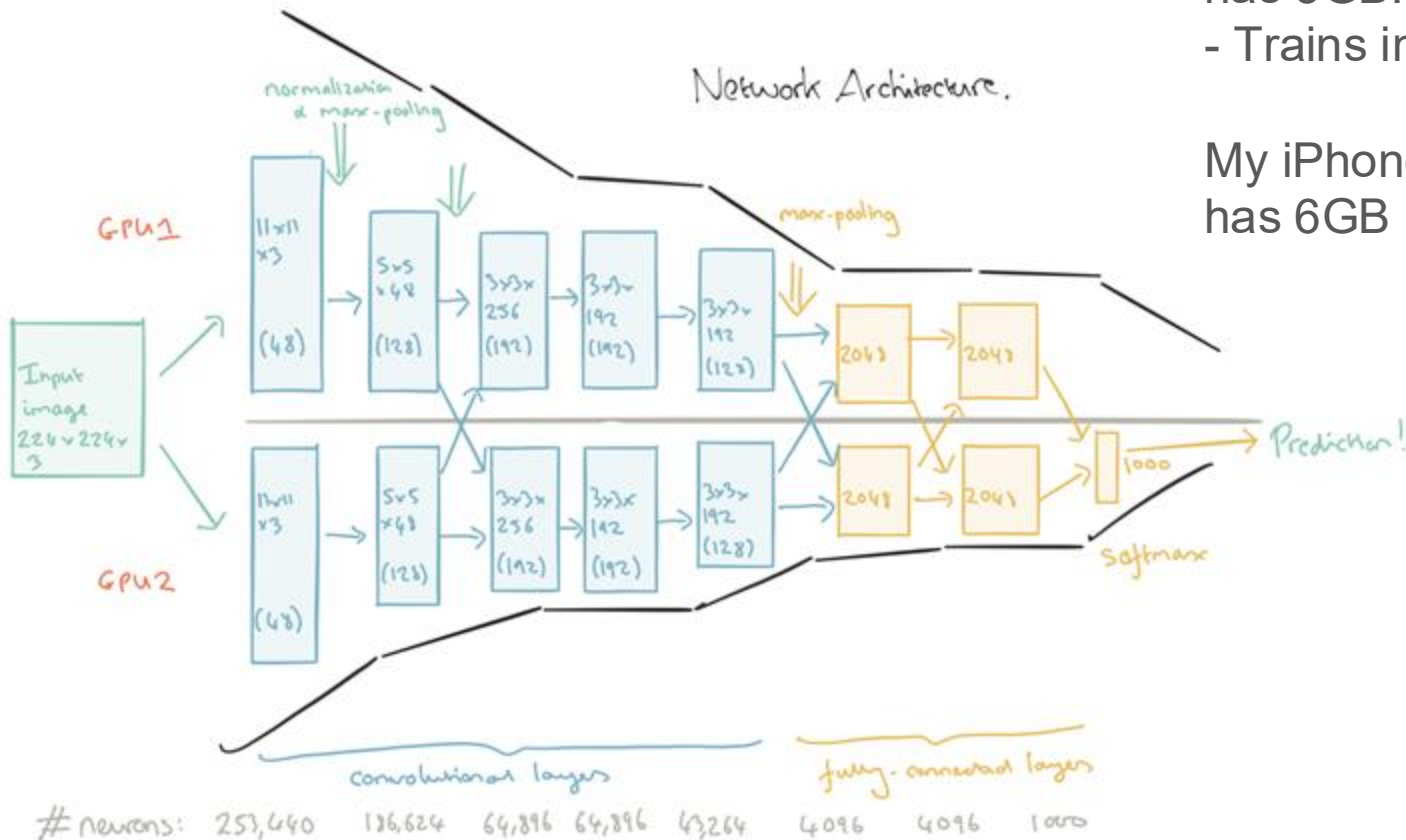


Figure 2: An illustration of the architecture of our CNN, explicitly showing the delineation of responsibilities between the two GPUs. One GPU runs the layer-parts at the top of the figure while the other runs the layer-parts at the bottom. The GPUs communicate only at certain layers. The network's input is 150,528-dimensional, and the number of neurons in the network's remaining layers is given by 253,440–186,624–64,896–64,896–43,264–4096–4096–1000.

Network Architecture (annotated)

GTX 580 GPU only
has 3GB!
- Trains in ~5 days

My iPhone's GPU
has 6GB ☺



Learned Convolutional Kernels

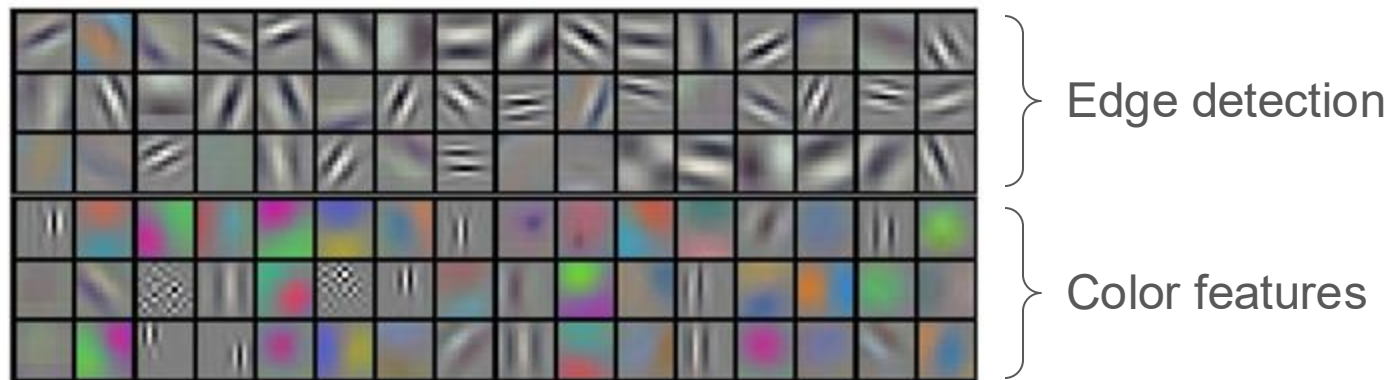


Figure 3: 96 convolutional kernels of size $11 \times 11 \times 3$ learned by the first convolutional layer on the $224 \times 224 \times 3$ input images. The top 48 kernels were learned on GPU 1 while the bottom 48 kernels were learned on GPU 2. See Section 6.1 for details.